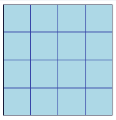
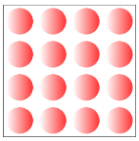
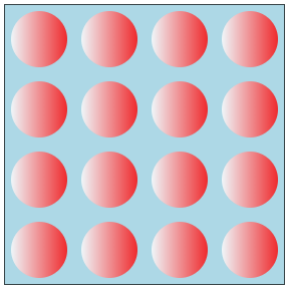
HTML5- SVG pattern

The <pattern> element is used to create a graphic that is redrawn at repeated x and y coordinate intervals, to cover an area. All SVG patterns are defined within a <defs> element. The <pattern> element has a required id attribute which identifies the pattern. The graphic/image then points to the pattern to use. Then, inside the <pattern> element, we put one or more elements that will be used as the fill pattern.

```
<svg width="200" height="200">
  <defs>
    <linearGradient id="grad1">
      <stop offset="0%" stop-color="white" />
      <stop offset="100%" stop-color="red" />
    </linearGradient>
    <pattern id="patt2" x="0" y="0" width="25%" height="0.25">
      <rect x="0" y="0" width="50" height="50" fill="lightblue"
        stroke="darkblue"/>
      <!--<circle cx="25" cy="25" r="20" fill="url(#grad1)" fill-
        opacity="0.8" />-->
    </pattern>
  </defs>
```







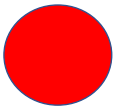
HTML5- SVG Animation

SVG elements can be animated. SVG has four animation elements which sets or animates SVG graphics: <set>, <animate>, <animateTransform>, <animateMotion>, to 50:

- The **<set>** element sets the value of an attribute for a specified duration.

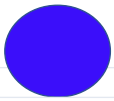
In below example, the red circle that starts with a radius of 25, then after 3 seconds, the radius will be set to 50:

```
<svg width="200" height="100">
  <circle cx="50" cy="50" r="25" style="fill:red;">
  <set attributeName="r" to="50" begin="3s" />
</svg>
```



- The **<animate>** element animates an attribute of an element. It should be nested inside the target element.

```
<svg width="100%" height="200" >
 <circle cx="150" cy="100" r="50" style="fill:rgb(21, 0, 255);">
  <animate attributeName="cx"begin="0s"dur="10s"from="50"to="100%"repeatCount="indefinite" fill="freeze"/>
 </circle>
</svg>
```



- The `fill="freeze"` attribute defines that the animation should freeze when it comes to its final position

HTML5- SVG Animation

SVG **<animateTransform>** animates the transform attribute on the target element. It should be nested inside the target element.

```
<svg width="200" height="180">
  <rect x="30" y="30" height="110" width="110" style="stroke:green;fill:red">
    <animateTransform attributeName="transform" begin="0s" dur="10s" type="rotate" from="0 85 85"
        to="360 85 85" repeatCount="indefinite" />
  </rect>
</svg>
```

- The **<animateMotion>** element sets how an element moves along a motion path. It should be nested inside the target element.

```
<svg width="100%" height="150" stroke="green">
 <rect x="45" y="18" width="155" height="45" style="stroke:green;fill:none;">
  <animateMotion path="M0,0 q60,100 100,0 q60,-20 100,0" begin="0s" dur="10s" repeatCount="indefinite"/>
 </rect>
  <text x="50" y="50" style="font-family:Verdana;font-size:32;color:#1b1b1b">It's SVG!
  <animateMotion path="M0,0 q60,100 100,0 q60,-20 100,0" begin="0s" dur="10s" repeatCount="indefinite"/>
</text>
```

It's SVG!